Enhanced dissipation for passive scalars

10 November 2026

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1 Overview

Discuss on Zulip: #overview, topic “Overview”.

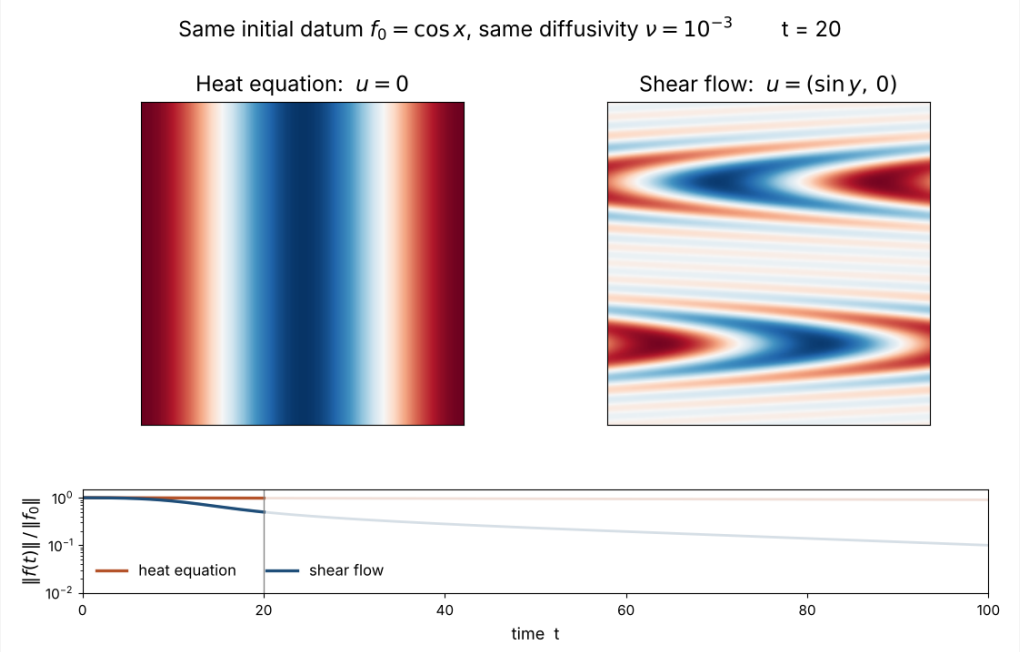
A passive scalar is a quantity carried by a fluid without acting on it, such as temperature, a dye in water or a pollutant in the air. If f is its concentration, u the velocity of an incompressible flow, and $\nu > 0$ the molecular diffusivity, then f solves the advection–diffusion equation

$$\partial_t f + u \cdot \nabla f = \nu \Delta f, \qquad \nabla \cdot u = 0.$$

Diffusion alone dissipates the scalar on a time scale of order $1/\nu$, which is very long when ν is small. The flow, on the other hand, dissipates nothing: it only rearranges the scalar. In doing so, it can transfer large-scale structures to small scales, for instance by stretching the scalar into thin filaments, and at small scales diffusion is much more efficient. The result is that f decays on a time scale much shorter than $1/\nu$: this is *enhanced dissipation*.

How, and by how much, does an incompressible velocity field speed up the dissipation of a diffusing scalar?

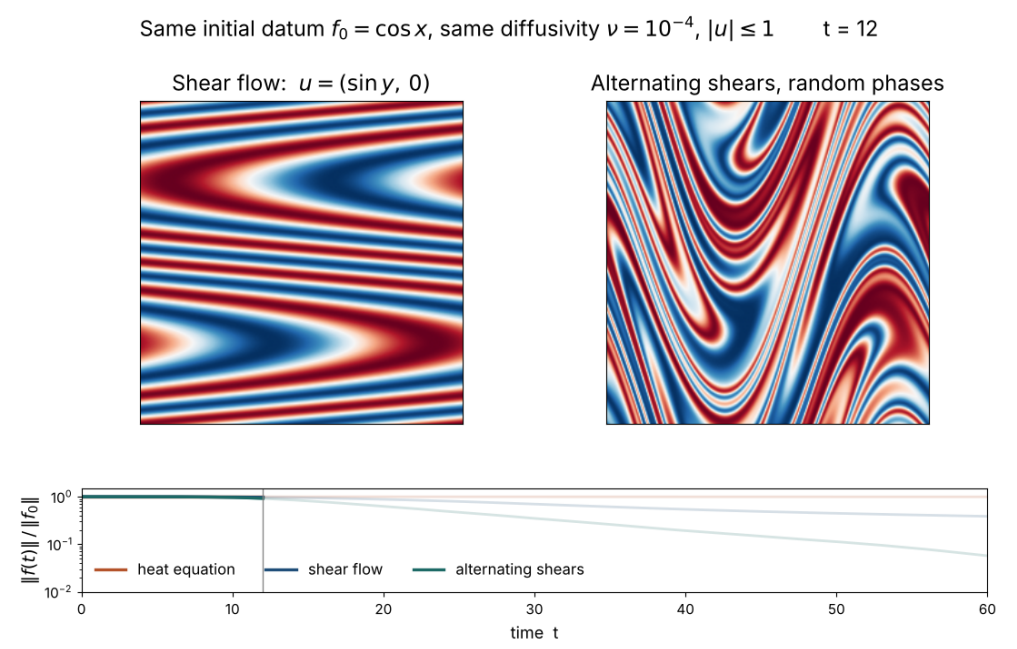
The two videos below show the phenomenon in the simplest setting, a periodic box in two dimensions. In the first, a steady shear flow significantly accelerates the dissipation of vertical stripes, orthogonal to its streamlines.



Video. Left: the heat equation. Right: the same initial datum carried by the shear flow $u = (\sin y, 0)$, with the same diffusivity. Bottom: the size of the two solutions in time, on a logarithmic scale. [numerics]
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-heat-vs-shear`.

The second video compares two flows of the same size. A steady shear stretches the scalar only along its own streamlines. Shears that alternate in direction, with random phases (the model of Pierrehumbert), stretch and fold the scalar, which then dissipates much faster and more uniformly in space.

Quantifying the time scale of enhanced dissipation is the subject of this wiki. Laminar flows, such as shear flows, typically dissipate the scalar on time scales of order ν^{-a} with $0 < a < 1$,



Video. Left: the shear flow $u = (\sin y, 0)$. Right: shears that alternate, horizontal and vertical, each acting for 3 units of time, with random phases. Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-4}$; both velocity fields have size at most 1. Bottom: the size of the solutions in time, on a logarithmic scale. [numerics]
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-alternating-shears`.

whereas *mixing* flows, such as the alternating shears, can dissipate it on time scales of order $|\log \nu|$. The questions are which time scales are possible, which flows achieve them, and how one proves it.

Mixing: the problem without diffusion. When $\nu = 0$ the equation is pure transport: the flow conserves every L^p norm of f and dissipates nothing. It can still *mix*, that is, move the scalar to ever smaller scales, so that f converges weakly to its mean while its L^2 norm stays constant. The speed of mixing can be measured by the decay of a negative Sobolev norm, such as $\|f\|_{H^{-1}}$. Mixing is a field of its own, with its own questions and methods; this wiki covers only its relation to enhanced dissipation.

Beyond passive scalars. The same mechanism, acting on the vorticity of a two-dimensional fluid, helps keep shear flows stable at large Reynolds numbers, against perturbations that are small enough in terms of the Reynolds number; the page Why it matters expands on this.

The setting, made precise. In most of the wiki, the domains without boundary are the torus $\mathbb{T}^2$ and the periodic strip $\mathbb{T} \times \mathbb{R}$; some problems are set in the whole plane $\mathbb{R}^2$. The domains with boundary are the disk and the periodic channel $\mathbb{T} \times [0, 1]$; the boundary conditions are stated where they are used. The initial datum f_0 is in L^2 and has mean zero; for shear and circular flows, it also has zero average along each streamline, since the flow does not act on that average. The size of a solution is its L^2 norm. One definition of

the dissipation time of a flow is the time it needs to reduce every initial datum by a fixed factor, say one half; the flow enhances dissipation when this time is much shorter than $1/\nu$. Definitions and notation are collected in Notation.

1.1 Map of the wiki

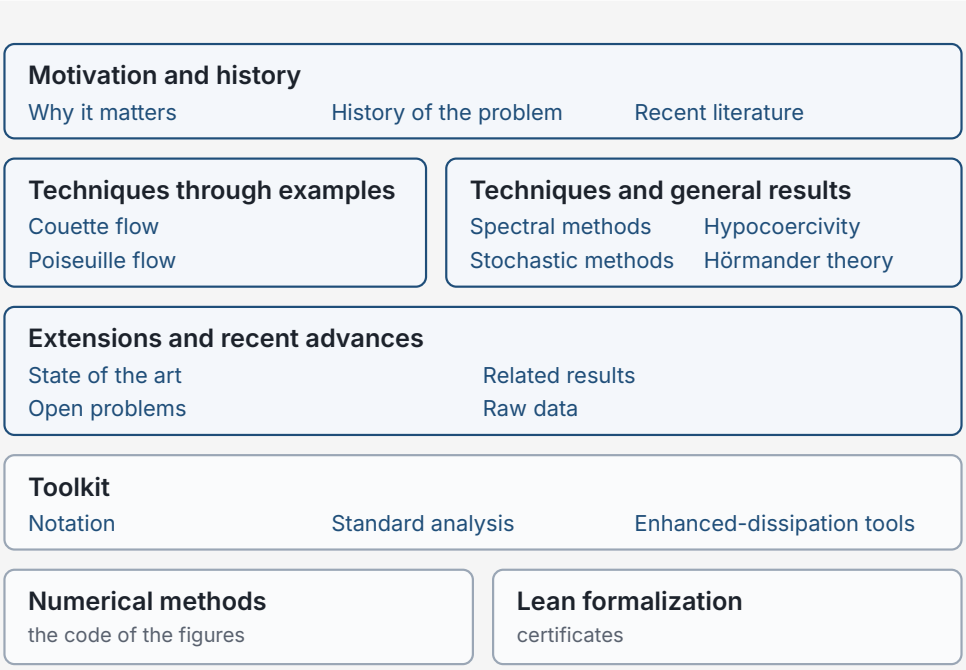


Diagram. The parts of the wiki; click a name to open its page.

- *Motivation and history*: why the problem matters and where it comes from.
- *Techniques through examples*: enhanced dissipation for two flows, Couette and Poiseuille, proved once with each method, so that the methods can be compared on the same example.
- *Techniques and general results*: the general theorem behind each method, with its proof, for spectral methods, hypocoercivity, stochastic methods and Hörmander theory.
- *Extensions and recent advances*: the state of the art, related results and open problems.
- *Toolkit*: the notation and the standard analysis that the proofs use.
- *Numerical methods*: the codes behind the pictures.
- *Lean formalization*: the statements checked by a proof assistant.
- *About this wiki*: how to contribute and who did what.

Part I
Motivation and history

2 Why it matters

Page team: to be named · Status: skeleton

Discuss on Zulip: #overview, topic “Why it matters”.

[Why the problem matters, in mathematics and in applications, written as the introduction of a paper.]

3 History of the problem

Page team: to be named · Status: skeleton
Discuss on Zulip: #overview, topic “History of the problem”.

[A short account of the progress; then the history technique by technique.]

3.1 Origins

[Before the mathematical theory.]

3.2 Spectral methods

[Contributions and dates; the general theory is on Spectral methods.]

3.3 Hypocoercivity

[Contributions and dates; the general theory is on Hypocoercivity.]

3.4 Stochastic methods

[Contributions and dates; the general theory is on Stochastic methods.]

3.5 Hörmander theory

[Contributions and dates; the general theory is on Hörmander theory.]

4 Recent literature

Page team: to be named · Status: skeleton
Discuss on Zulip: #overview, topic “Recent literature”.

[The latest papers, with what each contributes; their results are collected on State of the art and Related results.]

Part II
Techniques through examples

5 Couette flow

Page team: to be named · Status: skeleton
Discuss on Zulip: #examples, topic “Couette flow”.

[Lead: what this example shows, in three lines.]

5.1 Setting

[The flow $u = (y, 0)$ on $\mathbb{T} \times \mathbb{R}$: the equation, the Fourier modes, what decays.]

5.2 The result

Theorem 1 (enhanced dissipation for Couette flow)**.** *[The statement.]*

Lean: not yet formalized.

Idea. [The key mechanism in a few lines.]

Proof architecture. [The proof as a short chain of steps, and its diagram.]

5.3 One example, every technique

Each proof below is written for this example only, to show the mechanism; the general results, with complete proofs, are in the part *Techniques and general results*.

5.3.1 Explicit solution

Proof by the explicit solution. [The proof.] □

5.3.2 Spectral methods

Proof by spectral methods. [The proof.] □

General theory: Spectral methods.

5.3.3 Hypocoercivity

Proof by hypocoercivity. [The proof.] □

General theory: Hypocoercivity.

5.3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

General theory: Stochastic methods.

5.3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

General theory: Hörmander theory.

5.4 What this example teaches

[What the example shows, and what the general theory keeps of it.]

6 Poiseuille flow

Page team: to be named · Status: skeleton
Discuss on Zulip: #examples, topic "Poiseuille flow".

[Lead: what changes compared with Couette flow, and why.]

6.1 Setting

[The flow, the domain, the critical point, what decays.]

6.2 The result

Theorem 2 (enhanced dissipation for Poiseuille flow)**.** *[The statement.]*

Lean: not yet formalized.

Idea. [The key mechanism in a few lines.]

Proof architecture. [The proof as a short chain of steps, and its diagram.]

6.3 What changes compared with Couette flow

For each technique: what carries over from Couette flow, what is lost, and the proof for this example.

6.3.1 Explicit solution

[What is lost.]

6.3.2 Spectral methods

Proof by spectral methods. [The proof.] □

6.3.3 Hypocoercivity

Proof by hypocoercivity. [The proof.] □

6.3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

6.3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

Part III
Techniques and general results

7 Spectral methods

Page team: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Spectral methods".

[Lead: what the method gives, and under which hypotheses.]

7.1 The method

[The key mechanism, in a few lines.]
Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

7.2 General results

[The theorems, each written as a result (statement, Lean banner, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

7.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

8 Hypocoercivity

Page team: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Hypocoercivity".

[Lead: what the method gives, and under which hypotheses.]

8.1 The method

[The key mechanism, in a few lines.]

Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

8.2 General results

[The theorems, each written as a result (statement, Lean banner, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

8.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

9 Stochastic methods

Page team: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Stochastic methods".

[Lead: what the method gives, and under which hypotheses.]

9.1 The method

[The key mechanism, in a few lines.]

Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

9.2 General results

[The theorems, each written as a result (statement, Lean banner, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

9.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

10 Hörmander theory

Page team: to be named · Status: skeleton
Discuss on Zulip: #techniques, topic "Hörmander theory".

[Lead: what the method gives, and under which hypotheses.]

10.1 The method

[The key mechanism, in a few lines.]

Worked example: Couette flow. What changes with a critical point: Poiseuille flow.

10.2 General results

[The theorems, each written as a result (statement, Lean banner, idea, proof architecture, complete proof; see How to contribute); lemmas of general use go to the Toolkit *(Standard analysis, Enhanced-dissipation tools).]*

10.3 Limitations and comparison

[What the method does not give, and how it compares with the other techniques.]

Part IV
Extensions and recent advances

11 State of the art

Page team: to be named · Status: skeleton
Discuss on Zulip: #advances, topic "State of the art".

[The strongest results available to date, in one table.]

Flow class	Domain	Dissipation time	Sharp	Source	Label
—	—	—	—	—	—

12 Related results

Page team: to be named · Status: skeleton
Discuss on Zulip: #advances, topic "Related results".

Results related to enhanced dissipation and not yet part of the established theory. Each entry gives the statement, a label ([published], [preprint, to be verified] or [raw, unchecked]), the date it was added, its source, and where it would go in the established theory.

12.1 Entries

[First entries.]

12.2 How an entry moves

1. An entry keeps its label until its status changes: raw data checked and posted becomes [preprint, to be verified], a refereed result [published].
2. The editorial board moves established results into the part *Techniques and general results*, or into a new page, with a synthesis proof.

13 Open problems

Page team: to be named · Status: skeleton
Discuss on Zulip: #advances, topic “Open problems”.

Open problems, each with what is known, the expected answer, and the main difficulty.

13.1 Problems

[First problems.]

14 Raw data

Page team: to be named · Status: skeleton
Discuss on Zulip: #advances, topic “Raw data”.

AI-generated proofs, numerical observations and sketches, labelled [raw, unchecked] until a human has checked them. AI-generated text is quoted verbatim, with its source and date; how an entry moves on is explained on Related results.

14.1 Entries

[First entries.]

Part V
Toolkit

15 Notation

Page team: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic “Notation”.

The notation and the definitions used throughout the wiki.

15.1 Notation

[The scalar, the velocity field, the domains, the Fourier modes, the norms.]

15.2 Definitions

[Dissipation time, enhanced dissipation, and the related notions used in the wiki.]

16 Standard analysis

Page team: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic “Standard analysis”.

Classical facts of analysis used by the proofs. A statement comes to the Toolkit when several proofs use it, or when it holds in more generality than one result; a step used by one proof only is a claim inside that proof. Each statement is given once, with its proof or a reference, and the proofs elsewhere link to it.

16.1 Inequalities

[Poincaré and interpolation inequalities, and the other inequalities the proofs use.]

16.2 Functional analysis

[Semigroups, spectral theory, and the other facts the proofs use.]

17 Enhanced-dissipation tools

Page team: to be named · Status: skeleton
Discuss on Zulip: #toolkit, topic "Enhanced-dissipation tools".

Tools specific to enhanced dissipation. A statement comes to the Toolkit when several proofs use it, or when it holds in more generality than one result; a step used by one proof only is a claim inside that proof. Each statement is given once, with its proof or a reference, and the proofs elsewhere link to it.

17.1 Tools

[For instance spectral-gap and resolvent lemmas, and the link between mixing and enhanced dissipation.]

18 Numerical methods

Page team: to be named · Status: skeleton, one record
Discuss on Zulip: #numerics, topic "Numerical methods".

Every computed picture and video of this wiki has a record on this page: what was computed, how, with which parameters and checks, and the code that produced it.

18.1 How a record works

- A record is a section with an anchor `num-...`; the picture or video links to it with the parameter `record` of Media.
- It gives the equation, the parameters, the method, the checks, the code and how to run it, and its status.
- The code is in the folder `numerics/` of the repository of the wiki, one folder per record, with the output of its checks.

18.2 Records

18.2.1 Heat versus shear on the torus

Produces the video on the Overview (`heat-vs-shear.mp4`) and the check figure below

Equation $\partial_t f + \sin y\, \partial_x f = \nu \Delta f$ on $[0, 2\pi)^2$, periodic, and the heat equation $\partial_t f = \nu \Delta f$; same datum $f_0 = \cos x$

Parameters $\nu = 10^{-3}$, times $0 \le t \le 100$; video of 20 s at 24 frames per second, time running uniformly (5 time units per second), last frame held 1.5 s

Method one Fourier mode in x: $f = \mathrm{Re}(g(t,y)e^{ix})$ with $\partial_t g = -i \sin y\, g + \nu(\partial_y^2 - 1)g$; 512 Fourier modes in y; Strang splitting with time step 0.01, transport and diffusion steps solved exactly; heat equation solved exactly

Checks with 1024 modes and time step 0.005 the norms agree to a relative $6 \cdot 10^{-8}$; the energy balance $\|g(t)\|^2 + 2\nu \int_0^t \|(\partial_y g, g)\|^2 ds = \|g(0)\|^2$ holds to $3 \cdot 10^{-7}$; decay rate of the norm on $[50, 100]$: 0.0167, against $\nu^{1/2} \approx 0.0316$

Code numerics/heat-vs-shear/heat_vs_shear.py in the repository, also below (Python 3, NumPy, Matplotlib, ffmpeg); `python3 heat_vs_shear.py -out ../../wiki/media`, about a minute

Status computed by Claude (AI) on 8 November 2026; not yet checked by a human [demo text]

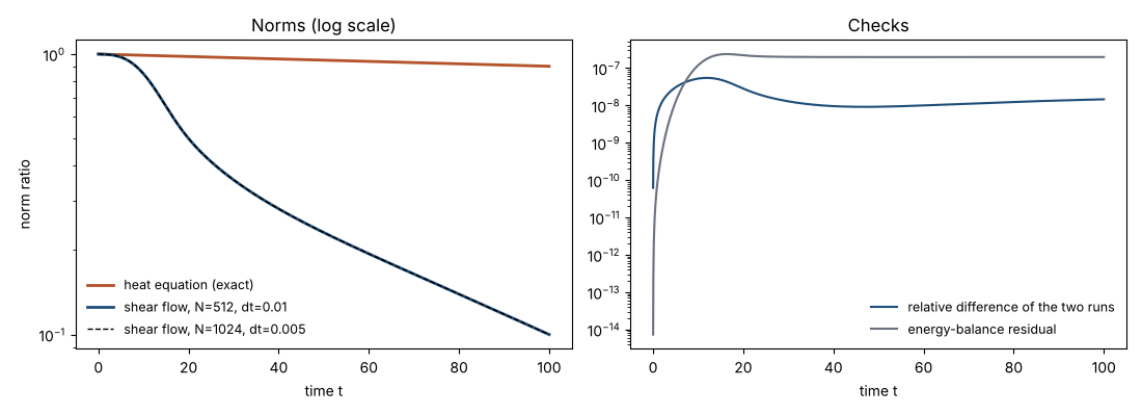


Figure. Left: the norms of the two runs and of the heat equation. Right: the relative difference of the two runs and the energy-balance residual.

Code.

```
#!/usr/bin/env python3
"""Heat equation versus advection-diffusion by a shear flow on the torus [0, 2pi)^2.

    d_t f + sin(y) d_x f = nu (d_xx + d_yy) f,      f(0, x, y) = cos x,

compared with the heat equation (the same without the transport term), whose solution is
exp(-nu t) cos x. The initial datum has zero average in x. It is a single Fourier mode in
    x,
so f(t, x, y) = Re(g(t, y) e^{ix}) with

    d_t g = -i sin(y) g + nu (d_yy - 1) g,      g(0, y) = 1,

solved by a Fourier method in y with Strang splitting: the transport step is the exact
multiplication by exp(-i sin(y) dt), the diffusion step is exact in Fourier variables.
```

```
The L2 norm reported is ||f(t)|| / ||f(0)||, that is the root mean square of |g(t, .)|.

Checks (written to heat-vs-shear-check.txt and heat-vs-shear-check.png):
  * the run is repeated with twice the Fourier modes and half the time step, and the norms
    are compared;
  * energy balance: ||g(t)||^2 + 2 nu int_0^t ||(d_y g, g)||^2 ds = ||g(0)||^2 (means over
    y).

The video lasts --duration seconds; frame s in [0, 1] shows t = T (a s + (1 - a) s^2), a =
    --slow (default 1:
time runs uniformly; a < 1 slows down the start); the last frame is held --hold seconds.

Outputs, in --out: heat-vs-shear.mp4 (video), heat-vs-shear-poster.png (one frame of it),
heat-vs-shear-check.png and heat-vs-shear-check.txt. Needs NumPy, Matplotlib and ffmpeg
(with the libx264 encoder). Run:  python3 heat_vs_shear.py --out ../../wiki/media

Drafted by Claude (AI) on 8 November 2026 for the wiki "Enhanced dissipation for passive
scalars"; record: page Numerical methods, section num-heat-vs-shear.
"""
import argparse
import os
import shutil
import subprocess
import tempfile

import numpy as np
import matplotlib

matplotlib.use("Agg")
import matplotlib.pyplot as plt

HEAT_COLOR = "#b4532a"
SHEAR_COLOR = "#1f4e79"


def solve(nu, T, N, dt, frame_times=()):
    """Times, norms ||f(t)||/||f(0)||, energy-balance residuals, and g(t, y) at
    frame_times."""
    y = 2 * np.pi * np.arange(N) / N
    eta = np.fft.fftfreq(N, d=1.0 / N)              # integer wavenumbers in y
    symbol = eta ** 2 + 1.0                          # symbol of -(d_yy - 1): mode k = 1
    in x
    half = np.exp(-nu * symbol * dt / 2)
    transport = np.exp(-1j * np.sin(y) * dt)
    ghat = np.fft.fft(np.ones(N, dtype=complex))
    steps = int(round(T / dt))
    times = dt * np.arange(steps + 1)
    sq = np.empty(steps + 1)                         # mean over y of |g|^2
    dissip = np.empty(steps + 1)                     # mean over y of |d_y g|^2 + |g|^2
    frames, targets, j = [], list(frame_times), 0
    for n in range(steps + 1):
        power = np.abs(ghat) ** 2 / N ** 2           # Parseval: means over y
        sq[n] = power.sum()
        dissip[n] = (symbol * power).sum()
        while j < len(targets) and targets[j] <= times[n] + dt / 2:
            frames.append(np.fft.ifft(ghat))
            j += 1
        if n == steps:
            break
        ghat *= half
        ghat = np.fft.fft(np.fft.ifft(ghat) * transport) * half
    integral = np.concatenate([[0.0], np.cumsum((dissip[1:] + dissip[:-1]) / 2) * dt])
```

```
    residual = np.abs(sq + 2 * nu * integral - 1.0)
    return times, np.sqrt(sq), residual, frames, y


def field(g, nx):
    """f(x, y) = Re(g(y) e^{ix}) on an nx-by-nx grid (rows: y, columns: x)."""
    step = max(1, len(g) // nx)
    x = 2 * np.pi * np.arange(nx) / nx
    return np.real(g[::step][:, None] * np.exp(1j * x)[None, :])


def draw_frame(path, t, g, nu, times, norms, heat, nx, ymin):
    x = 2 * np.pi * np.arange(nx) / nx
    f_heat = np.exp(-nu * t) * np.cos(x)[None, :] * np.ones((nx, 1))
    f_shear = field(g, nx)
    fig = plt.figure(figsize=(11.2, 7.2), dpi=100)
    gs = fig.add_gridspec(2, 2, height_ratios=[3.6, 1.15], hspace=0.3, wspace=0.06,
                          left=0.08, right=0.97, top=0.845, bottom=0.095)
    for col, (data, title) in enumerate([(f_heat, "Heat equation:  $u = 0$"),
                                         (f_shear, r"Shear flow:  $u = (\sin y,\, 0)$")]):
        ax = fig.add_subplot(gs[0, col])
        ax.imshow(data, origin="lower", extent=[0, 2 * np.pi, 0, 2 * np.pi], cmap="RdBu_r"
    ,
                  vmin=-1, vmax=1, interpolation="bilinear")
        ax.set_title(title, fontsize=17, pad=8)
        ax.set_xticks([]); ax.set_yticks([])
        ax.set_aspect("equal")
    fig.suptitle(r"Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-3}$"
                 + "        t = %d" % round(t), fontsize=17, y=0.975)
    ax = fig.add_subplot(gs[1, :])
    k = np.searchsorted(times, t, side="right")
    ax.semilogy(times, heat, color=HEAT_COLOR, alpha=0.18, lw=2)
    ax.semilogy(times, norms, color=SHEAR_COLOR, alpha=0.18, lw=2)
    ax.semilogy(times[:k], heat[:k], color=HEAT_COLOR, lw=2.4, label="heat equation")
    ax.semilogy(times[:k], norms[:k], color=SHEAR_COLOR, lw=2.4, label="shear flow")
    ax.axvline(t, color="#888888", lw=1)
    ax.set_xlim(0, times[-1]); ax.set_ylim(ymin, 1.5)
    ax.set_yticks([10.0 ** e for e in range(int(np.log10(ymin)), 1)])
    ax.set_xlabel("time  t", fontsize=13)
    ax.set_ylabel(r"$\|f(t)\|\,/\,\|f_0\|$", fontsize=14)
    ax.tick_params(labelsize=11)
    ax.legend(loc="lower left", fontsize=12, frameon=False, ncol=2)
    fig.savefig(path)
    plt.close(fig)


def main():
    ap = argparse.ArgumentParser(description=__doc__.split("\n")[0])
    ap.add_argument("--out", default=".", help="folder for the video and the figures")
    ap.add_argument("--nu", type=float, default=1e-3)
    ap.add_argument("--T", type=float, default=100.0)
    ap.add_argument("--N", type=int, default=512, help="Fourier modes in y")
    ap.add_argument("--dt", type=float, default=0.01)
    ap.add_argument("--duration", type=float, default=20.0, help="length of the video in
    seconds")
    ap.add_argument("--fps", type=int, default=24)
    ap.add_argument("--slow", type=float, default=1.0, help="speed of time at the start,
    relative to the mean (1: uniform)")
    ap.add_argument("--hold", type=float, default=1.5, help="seconds the last frame is
    held")
    ap.add_argument("--nx", type=int, default=256, help="pixels per side of each panel")
    a = ap.parse_args()
```

```
    os.makedirs(a.out, exist_ok=True)
    s = np.linspace(0, 1, int(round(a.duration * a.fps)))
    frame_times = a.T * (a.slow * s + (1 - a.slow) * s ** 2)
    times, norms, residual, frames, y = solve(a.nu, a.T, a.N, a.dt, frame_times)
    heat = np.exp(-a.nu * times)
    # check: the same run with N doubled and dt halved
    times2, norms2, residual2, _, _ = solve(a.nu, a.T, 2 * a.N, a.dt / 2)
    diff = np.abs(norms2[::2] - norms) / norms2[::2]
    ymin = 10 ** np.floor(np.log10(norms.min()))
    with open(os.path.join(a.out, "heat-vs-shear-check.txt"), "w") as fh:
        fh.write("nu=%g T=%g N=%d dt=%g frames=%d fps=%d\n" % (a.nu, a.T, a.N, a.dt, len(
    frame_times), a.fps))
        fh.write("final norm: shear %.4e, heat %.4e\n" % (norms[-1], heat[-1]))
        fh.write("max relative difference with (2N, dt/2): %.2e\n" % diff.max())
        fh.write("max energy-balance residual: %.2e (2N, dt/2: %.2e)\n" % (residual.max(),
     residual2.max()))
        fh.write("decay rate of the shear norm on [T/2, T]: %.4f; nu^(1/2) = %.4f\n" % (
            np.log(norms[len(norms) // 2] / norms[-1]) / (times[-1] - times[len(times) //
    2]), np.sqrt(a.nu)))
    # check figure
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4), dpi=100)
    ax1.semilogy(times, heat, color=HEAT_COLOR, lw=2, label="heat equation (exact)")
    ax1.semilogy(times, norms, color=SHEAR_COLOR, lw=2, label="shear flow, N=%d, dt=%g" %
    (a.N, a.dt))
    ax1.semilogy(times2, norms2, color="k", lw=1, ls="--", label="shear flow, N=%d, dt=%g"
     % (2 * a.N, a.dt / 2))
    ax1.set_xlabel("time t"); ax1.set_ylabel("norm ratio"); ax1.legend(frameon=False,
    fontsize=9)
    ax1.set_title("Norms (log scale)")
    ax2.semilogy(times[1:], diff[1:] + 1e-16, color=SHEAR_COLOR, lw=1.5, label="relative
    difference of the two runs")
    ax2.semilogy(times[1:], residual[1:] + 1e-16, color="#6b7280", lw=1.5, label="energy-
    balance residual")
    ax2.set_xlabel("time t"); ax2.legend(frameon=False, fontsize=9); ax2.set_title("Checks
    ")
    fig.tight_layout()
    fig.savefig(os.path.join(a.out, "heat-vs-shear-check.png"))
    plt.close(fig)
    # video
    with tempfile.TemporaryDirectory() as tmp:
        for i, (t, g) in enumerate(zip(frame_times, frames)):
            draw_frame(os.path.join(tmp, "f%04d.png" % i), t, g, a.nu, times, norms, heat,
     a.nx, ymin)
        poster = min(range(len(frame_times)), key=lambda i: abs(frame_times[i] - 0.2 * a.T
    ))
        shutil.copyfile(os.path.join(tmp, "f%04d.png" % poster), os.path.join(a.out, "heat
    -vs-shear-poster.png"))
        subprocess.run(["ffmpeg", "-y", "-loglevel", "error", "-framerate", str(a.fps),
                        "-i", os.path.join(tmp, "f%04d.png"), "-vf", "tpad=stop_mode=clone
    :stop_duration=%g" % a.hold,
                        "-c:v", "libx264", "-pix_fmt", "yuv420p",
                        "-crf", "26", "-preset", "slow", "-movflags", "+faststart",
                        os.path.join(a.out, "heat-vs-shear.mp4")], check=True)
    print(open(os.path.join(a.out, "heat-vs-shear-check.txt")).read().strip())


if __name__ == "__main__":
    main()
```

18.2.2 Alternating shears with random phases

Produces the second video on the Overview (`alternating-shears.mp4`) and the check figure below

Equation $\partial_t f + u \cdot \nabla f = \nu \Delta f$ on $[0, 2\pi)^2$, periodic, with $f_0 = \cos x$ and the alternating shears of Pierrehumbert's model: $u = (\sin(y + \varphi_n), 0)$ for $6n \le t < 6n + 3$ and $u = (0, \sin(x + \psi_n))$ for $6n + 3 \le t < 6n + 6$, the phases independent and uniform on $[0, 2\pi)$; compared with the steady shear $u = (\sin y, 0)$ and with the heat equation. Both velocity fields have size at most 1.

Parameters $\nu = 10^{-4}$, each shear acting for a time 3, times $0 \le t \le 60$, phases from the random generator of NumPy with seed 2026; video of 20 s at 24 frames per second, time running uniformly (3 time units per second), last frame held 1.5 s

Method Fourier pseudo-spectral method on a 512×512 grid; Strang splitting with time step 0.0125, the transport steps (each row or column shifted, through its Fourier coefficients) and the diffusion steps solved exactly; the steady shear as in the record above

Checks with 1024×1024 modes and time step 0.00625 the norms agree to a relative $1.2 \cdot 10^{-6}$; the energy balance $\|f(t)\|^2 + 2\nu \int_0^t \|\nabla f\|^2 ds = \|f_0\|^2$ holds to $3.5 \cdot 10^{-6}$; time for the norm to halve, for $\nu = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$: alternating shears 8.8, 16.3, 23.9, 31.7 (the same with twice the modes), an increase of about 7.6 for each factor 10 in ν; steady shear 8.6, 20.0, 43.9, 94.9; heat equation $\ln 2/\nu$

Context Exponential mixing for randomly shifted alternating shears is proved by Blumenthal, Coti Zelati and Gvalani (2023), and dissipation on the time scale $|\log \nu|$, with high probability, by Cooperman, Iyer and Son (2025); whether the parameters used here satisfy their hypotheses has not been checked

Code numerics/alternating-shears/alternating_shears.py in the repository (Python 3, NumPy, SciPy, Matplotlib, ffmpeg); `python3 alternating_shears.py -out ../../wiki/media`, about 15 minutes

Status computed by Claude (AI) on 8 November 2026; not yet checked by a human [demo text]

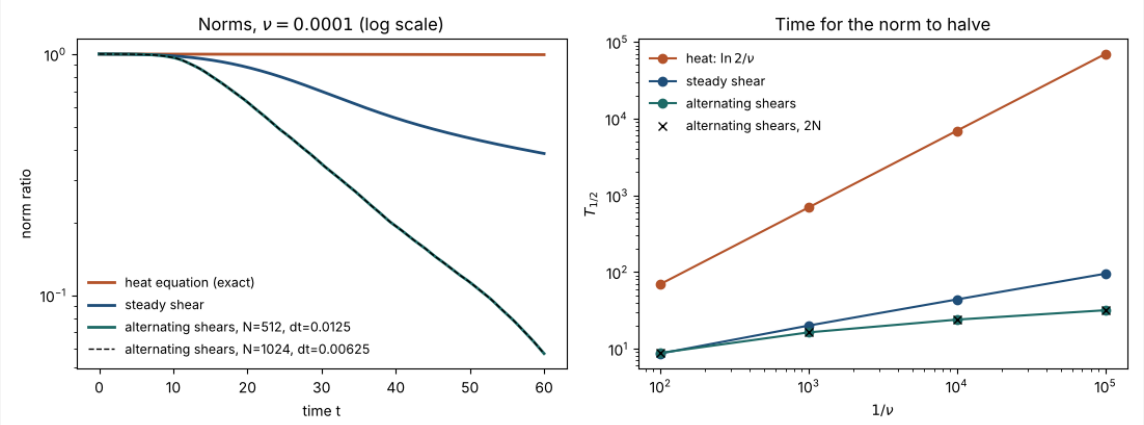


Figure. Left: the norms for $\nu = 10^{-4}$, with two resolutions for the alternating shears. Right: the time for the norm to halve, against $1/\nu$, logarithmic scales.

18.3 Solvers

[The methods used across the records, with their strengths and limits.]

19 Lean formalization

Page team: to be named · Status: skeleton
Discuss on Zulip: #lean, topic “Lean formalization”.

A result of this wiki can carry a machine-checked proof in Lean, its *certificate*. A small banner under each theorem says where it stands. A result not yet formalized:
Lean: not yet formalized.

A formalized result, with a link to its certificate:
Lean: formalized.

19.1 Certificates

Result	Status	Certificate	Lean and Mathlib versions
—	—	—	—

19.2 How to add a certificate

[What a certificate contains (repository, commit, versions, the statement as formalized, the axioms used), and how it is checked before its banner turns green.]

Part VI
About this wiki

20 How to contribute

Page team: Michele Dolce · Status: proposal, to be agreed with the editorial board · Text: [demo text]
Discuss on Zulip: #contributing, topic “How to contribute”.

This wiki is written from the bottom up, the way a journal works: members of the community propose changes on the talk page of the page they change, proposals are discussed and reviewed there in the open, an associate editor decides, and accepted changes are written into the wiki. The roles, and who holds them, are on Contributors. [demo text]

20.1 Roles

- **Editorial board**: the scope and the structure of the wiki; appoints the associate editors; settles discussions that end without consensus.
- **Associate editors**: one or more per part, following the talk pages of its pages and its Zulip channel. They run the discussion of each proposal, ask members of the community for a review when it helps, and decide.
- **Community**: propose, review and discuss. Every member can endorse a new member. Editors can remove a member whose participation disturbs the work rather than helping it.

The first line of each page names its team. As the wiki grows, busy parts and pages get more associate editors; the roles stay the same. On the wiki, everyone with a role is a *member* (reads every page and writes on talk pages); the editorial board and the associate editors are also *editors* (edit articles and templates, and write accepted proposals into the pages); administrators give these rights.

20.2 Proposing a change

Every change to a page, from a typo to a new proof, is a *proposal*: a topic on the talk page of the page (tab *Discussion*), which members of the wiki open and discuss. There are two ways to open one.

- *One paragraph, a figure, the central question, or a part still to be written* (a typo, a sentence, a formula, a shorter argument, a caption, a shorter video, a missing proof): in the edition, the pencil button next to it shows its TeX or, for a part still to be written, an empty box (the link *Write this part* opens it too). Change the TeX, write the missing text, or describe the change you ask for (for a picture or a video, for instance), say why, and open the proposal: the wiki shows a new topic on the talk page, already filled in, to be saved. A text too long for a link is copied by the edition, to be pasted into the topic in the wiki.
- *Larger changes* (a section, a statement, a proof, a new section): the link *new proposal* under the title of the page in the wiki (*Propose a change* in the edition) opens a new topic with a short form. Download the TeX source of the page: link *TeX source* under the title of every page of the edition (with the figures, in a zip, when the page has some). It is a standard LaTeX article that compiles as it is. The TeX of the whole wiki (*TeX of the whole wiki*, on the Overview) is one LaTeX project, for working offline, in which every page is a file of the folder *pages*; such a file can be proposed in the same way. Change what you want and check that it compiles; then paste in the form the blocks you changed (a section, or a statement or a proof with its label), or the whole file: only what changes is applied, and there is no limit of length. Say what the proposal changes, why, and which sources it uses; for a long proposal, attach its PDF (upload it and link it in the form).

Those who are not yet members of the wiki write in the Zulip channel *contributing*, naming the page.

20.2.1 How a proposal is decided

1. The community discusses and reviews the proposal in its topic, with the buttons *Support*, *Oppose* and *Comment* of its box, which add a signed reply (the box counts the supports and the oppositions); replies can also be written by hand, signed with `~~~~`. The author may revise the proposed text in place; the history of the talk page keeps every version.
2. When the discussion has settled, an associate editor of the part decides, with the buttons *Accept* or *Decline* in the box at the top of the topic (shown to editors): the box changes status, and the reason they give is added as a signed reply. There is no deadline. If the discussion ends without consensus, the editorial board decides.
3. An editor writes the accepted proposal into the page with a script. The script converts the TeX to the wiki's format, refuses a paragraph that has changed since the proposal was made, shows exactly what changes on the page, and flags new citations and links. The edit summary links the topic and credits its author, and the box of the topic becomes *applied*, with a link to the change. Decided topics, applied or declined, are moved to the archive of the talk page (its subpage */Archive*), so that the talk page shows only the proposals under discussion or waiting to be applied.

Small corrections (a typo, a broken link, a missing reference) go the same way, and are usually accepted without discussion. Open and accepted proposals are listed in :Category:Open proposals and :Category:Accepted proposals. A proposal changes only the blocks it contains. A paper that is not yet on Literature comes with the proposal that cites it (see *Citations* in The TeX format), and is added to the reference catalogue when the proposal is applied.

20.3 Credits

Under the title of a section, a line in small type names the people who contributed to it, the most recent first: `{{Contributed by|Name Surname, Name Surname}}` on the line after the heading (in TeX, `\contributors{Name Surname, Name Surname}` after the section title). A relevant contribution cannot be measured objectively: the editors decide, at their discretion, who is listed, and add a new name at the beginning of the list. A long list shows the three most recent names, and a link opens the others. The history of each page keeps the record of every change and of its author.

20.4 The TeX format

The TeX source of a page is a standard LaTeX article: one file that compiles with pdflatex and nothing else (on Overleaf, or with TeXShop, TeXstudio, ...; typeset twice for the citations). Its preamble is the same for every page and is never changed; a contribution defines no macros, since the wiki has none. Sections are numbered in the PDF only; the website numbers the results within each page (Theorem 1, Lemma 2, ...).

- Statements: theorem, proposition, lemma, corollary, conjecture, definition, example, remark, claim. The optional argument is the name, the label is the anchor: `\begin{theorem}[name]\label{thm-name}`. Labels start with thm-, prp-, lem-, cor-, def- or clm-.
- After a statement: `\lean{}` (the Lean banner), the environments idea and architecture, and proofs with an optional title and a label, as in `\begin{proof}[Proof by hypocoercivity]\label{proof-name}`. A claim and its proof sit inside the proof that uses them.
- Formulas: `$...$`, `\[ ... \]` and align*; no numbered equations and no labels on equations; none of the commands that the wiki's renderer rejects (the script lists them).
- Citations: `\citet{Key}` gives *Author (year)* and `\citep{Key}` gives *(Author, year)*, with the keys of Literature; several papers at once: `\citep{Key1,Key2,Key3}`. The reference list at the end of each page is generated from the reference catalogue and contains only the papers that the page cites: a new citation is added to it when the proposal is applied. A paper that is not yet on Literature: give its full reference in the proposal, as `\bibitem{NewKey} Authors, title, journal (year)` in a `thebibliography` at the end of the TeX, with its DOI or arXiv number (from the pencil button: in the reason field); it is added to the catalogue and then to the page.
- Links: on the same page, `\hyperref[thm-name]{the theorem above}` rather than `\ref`, since the wiki has no numbers; to another page, `\wikilink{Page\#anchor}{text}`.
- Labels: `\heuristic{}`, `\numerics{}`, `\published{}`, `\preprint{}`, `\raw{}`.
- A new block goes after an existing one: put the line `%% after: Setting` (a section title or a label) just before it.

20.5 Anatomy of a result

A result is a statement followed by its levels of detail, each folded until the reader opens it:

Statement	in the simplest form, constants hidden
Lean	a one-line banner: formalized or not yet
Idea	the key mechanism in a few lines
Architecture	the proof as a short chain of steps, later with a diagram
Proof	the complete proof: first its sources (the papers it follows, cited by their key), then the steps; a statement used only in this proof is a *claim*, with its own folded proof inside

A statement that several proofs use, or that holds in more generality than one result, goes to the *Toolkit* (Standard analysis or Enhanced-dissipation tools) and is linked from the proofs. A short proof is written in one piece. The specimen below shows the boxes at work; open the folds to see the nesting.

20.5.1 Specimen

Theorem 3 (specimen)**.** *The statement, in its simplest form.*

Lean: not yet formalized.

Idea. The key mechanism in a few lines, and the heuristic scaling, labelled [heuristic].

Proof architecture. 1. Step 1 reduces the theorem to a claim.

2. Step 2 proves the claim.

3. Step 3 concludes with a lemma of the toolkit.

Proof. Sources. This proof follows ...
Step 1. The reduction.

Claim (specimen)**.** *A statement used only in this proof.*

Proof of the claim. The proof of the claim, folded inside the proof that uses it. □

Step 2. The conclusion, with a lemma of the toolkit. □

20.6 Labels

- [demo text]: text drafted by Claude (AI), until a member of the page team replaces or approves it.
- [heuristic]: a non-rigorous explanation, kept apart from what is proved.
- [numerics]: a computation, with its record on Numerical methods.
- [published], [preprint, to be verified], [raw, unchecked]: the status of an entry on Related results and Raw data: refereed; posted but not refereed; neither posted nor checked by a human (an AI-generated proof, a numerical observation, a sketch).
- Grey quote boxes: AI-generated text, quoted verbatim with its source and date.

20.7 Pictures, videos and numerics

A picture, a video or a diagram is one line in the wiki, the template Media; in TeX, `\media{video}{file}{caption}`. The files are kept in the folder `wiki/media` of the project; the edition shows them, the wiki shows a box with the caption. Every computed picture or video has a record on Numerical methods, with the method, the parameters, the checks and the code.

20.8 Lean

Under each theorem a one-line banner says whether it is formalized: grey while it is not, green with a link to the certificate once it is. Certificates are listed on Lean formalization.

20.9 Discussion and conduct

Proposals are discussed on the talk page of the page they change, so that the discussion stays next to the text and in its history. Zulip is the forum for the mathematics: questions and clarifications, ideas for new results and how they fit the established theory, and larger changes to the structure of the wiki, discussed there before they become proposals. It has one channel per part and one topic per question or idea; the link under the title of each page opens the topic named after the page, for questions about it.

Part	Zulip channel
Overview, Motivation and history	overview
Techniques through examples	examples
Techniques and general results	techniques
Extensions and recent advances	advances
Toolkit	toolkit
Numerical methods	numerics
Lean formalization	lean
About this wiki	contributing

Discuss the mathematics, not the people. Associate editors moderate the channel and the talk pages of their part, and can hide messages; editors can remove a member whose participation disturbs the work rather than helping it. The Zulip is endipa.zulipchat.com, by invitation; its administrators invite new members and create channels.

20.10 Formulas in the wiki

For editors who write the wiki's own format.

- Inline formulas need `display="inline"`, displayed ones `display="block"`; each formula is written on one line. The renderer rejects `\newcommand`, `\label`, `\eqref` and `\substack`. No formulas in headings.
- Citations link to the Literature page by the key of the reference catalogue: `[[Literature#BedrossianCotiZelati2017|Bedrossian and Coti Zelati (2017)]]`. The section *References* of each page is generated from the catalogue: add the key to its comment line rather than editing the list.
- To link to a result, use the `id` of its box as the anchor, as in `Couette flow#thm-couette`.

20.11 The edition

A static edition of this wiki, with the pictures, the videos and the TeX source of every page, is built from the wiki's text by a script and never edited by hand: endipa.github.io.

21 Contributors

Page team: editorial board · Status: to be filled
Discuss on Zulip: #contributing, topic “Contributors”.

The people who write and check this wiki, by role; the contributors of each section are also named under its title. How they work together is described in How to contribute.

21.1 Editorial board

Decides the scope and the structure of the wiki, appoints the associate editors, and settles a discussion that ends without consensus.
[Names and roles.]

21.2 Associate editors

Each part of the wiki has at least one associate editor, who follows the talk pages of its pages and its Zulip channel, runs the discussion of every proposal, asks members of the community for a review when it helps, and decides; accepted changes are written into the wiki by an editor. A busy part or page can have several.
[Names, with their parts.]

21.3 Community

Members of the wiki and of the Zulip, who propose, review and discuss changes. Every member can endorse a new member; editors can remove a member whose participation disturbs the work rather than helping it. Early-stage researchers (PhD students, and researchers up to two years after the PhD) can choose to be listed as such.
[Names.]

22 Provenance and AI use

Page team: Michele Dolce · Text: [demo text]
Discuss on Zulip: #contributing, topic “Provenance and AI use”.

This page records where each piece of the wiki comes from and who has checked what. The labels are explained on How to contribute. [demo text]

22.1 Rules

- Text drafted by Claude (AI) carries [demo text] until a member of the page team replaces or approves it.
- Arguments produced by AI systems appear only as verbatim quotations, with source and date, or as entries labelled [raw, unchecked] on Raw data.
- The first line of each page records its team, its status and the checks made. A machine check is recorded as such and never counts as a human check.
- Statements attributed to a paper give its theorem number; synthesis proofs say which papers they follow.
- Every accepted proposal is linked, in the edit summary, to the Zulip topic where it was discussed.

22.2 This version

On 8 November 2026 the wiki was reorganized into the parts drawn on the map, and its pages were reduced to a skeleton, to be filled by the editorial board and contributors. The first version (6 November 2026: Couette flow with three proofs, a hypocoercivity page, the toolkit, the history) is kept in the project folder, in `archive/2026-11-06-first-version`. The only computed content is the video on the Overview, with its record on Numerical methods, made by Claude (AI) and not yet checked by a human. The TeX source of each page is generated from the wiki by a script.

22.3 AI-generated documents

Document	Date	Contents	Checks recorded
none	—	—	—

22.4 How a result becomes checked

[To be agreed with the editorial board: how many readers a proof needs before it loses its label, and whether a page records who read which proof.]

23 Literature

Page team: to be named · Status: generated from the reference catalogue (references/catalog.json, checked 2026-11-05)
Discuss on Zulip: #contributing, topic "Literature".

The papers of the reference catalogue, one line each, grouped by topic. This page is generated by a script from the catalogue; a new paper usually arrives with the proposal that cites it, and an entry is corrected by a proposal on the talk page of this page (see How to contribute); the catalogue is updated first. Pages cite these entries by their key, for instance Bedrossian and Coti Zelati (2017). Preprints are listed with their arXiv identifier. The list is a working bibliography, not a survey of the field.

23.1 Orientation

1. M. Coti Zelati, G. Crippa, G. Iyer and A. L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay*, Notices of the American Mathematical Society **71** (2024), 593–604, arXiv:2308.00358.

2. A. L. Mazzucato, Y. Feng and C. Nobili, *Enhanced dissipation by advection and applications to PDEs*, Applied Mathematical Modelling **150** (2026), Part A, 116310, arXiv:2501.17695.

23.2 General mechanisms and rate limitations

1. P. Constantin, A. Kiselev, L. Ryzhik and A. Zlatoš, *Diffusion and mixing in fluid flow*, Annals of Mathematics **168** (2008), 643–674, arXiv:math/0509663.

2. W. Cooperman, G. Iyer, K. Rowan and S. Son, *Exponentially mixing flows with slow enhanced dissipation*, preprint (2025), arXiv:2507.21305.

3. M. Coti Zelati, M. G. Delgadino and T. M. Elgindi, *On the relation between enhanced dissipation timescales and mixing rates*, Communications on Pure and Applied Mathematics **73** (2020), 1205–1244, arXiv:1806.03258.

4. Y. Feng and G. Iyer, *Dissipation enhancement by mixing*, Nonlinearity **32** (2019), 1810–1851, arXiv:1806.03699.

5. C. Seis, *Bounds on the rate of enhanced dissipation*, Communications in Mathematical Physics **399** (2023), 2071–2081, arXiv:2003.08794.

6. D. Wei, *Diffusion and mixing in fluid flow via the resolvent estimate*, Science China Mathematics **64** (2021), 507–518, arXiv:1811.11904.

23.3 Smooth stationary shears

1. D. Albritton, R. Beekie and M. Novack, *Enhanced dissipation and Hörmander's hypoellipticity*, Journal of Functional Analysis **283** (2022), 109522, arXiv:2105.12308.

2. J. Bedrossian and M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*, Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204, arXiv:1510.08098.

3. M. Coti Zelati, *Stable mixing estimates in the infinite Péclet number limit*, Journal of Functional Analysis **279** (2020), 108562, arXiv:1909.01310.

4. M. Coti Zelati and T. D. Drivas, *A stochastic approach to enhanced diffusion*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834, arXiv:1911.09995.

5. M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.

23.4 Circular and Hamiltonian flows and Taylor dispersion

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